

International Cybersecurity and Digital Forensics Academy (ICDFA)



SBT-DF203 Lab 9 Report

WEP40 Wireless Packet Decryption and Aircrack Forensics

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Student ID:	2025-FWSD-11392
Course Title:	Basic Networking Skills for Digital Forensics
Course Code:	SBT-DF203
Lab Number:	Lab 9
Instructor:	Aminu Idris, AMCPN
Case/Lab Identifier:	SBT-DF203-Lab9-Abubakari Sadique Hamidu
Date:	25 September 2026

1. Source Materials

This practical aligns directly with the SBT-DF203 lecture series covering IEEE 802.11 wireless frame structures, WEP/RC4 stream cipher architecture, Initialization Vector (IV) reuse weaknesses, key recovery using Aircrack-ng, offline packet decryption via airdecap-ng, endpoint conversation reconstruction, and file carving.

2. Executive Summary / Lab Scenario

A historical wireless packet capture (file.xz) originating from the CodeGate CTF 2015 "Good Crypto" challenge was analyzed offline. The goal was to decompress and preserve the capture, audit 802.11 management, control, and data frames, demonstrate WEP40 structural vulnerabilities, recover the 40-bit key (A4:3D:F6:F3:74), decrypt the payload to produce file_working-dec, inventory layer 3 endpoints, and extract transferred HTTP objects and carved file artifacts.

3. Learning Outcomes Demonstrated

- Differentiated Management (Type 0), Control (Type 1), and Data (Type 2) 802.11 frames.
- Analyzed WEP40 framing (24-bit IV + 40-bit key) and identified IV collision patterns.
- Executed PTW key recovery against 15,477 IVs using aircrack-ng.
- Decrypted 15,477 WEP data packets offline using airdecap-ng.
- Extracted Ethernet endpoints, IP endpoints, and TCP conversations from decrypted streams.
- Extracted transferred HTTP objects and carved binary objects using foremost.

4. Legal, Ethical and Safety Rules Compliance

- Conducted analysis exclusively offline on an authorized historical CTF capture.
- No live wireless networks were targeted, intercepted, or probed.
- Original evidence files (file.xz) were preserved untouched and verified via SHA-256.

5. Tools and Environment Used

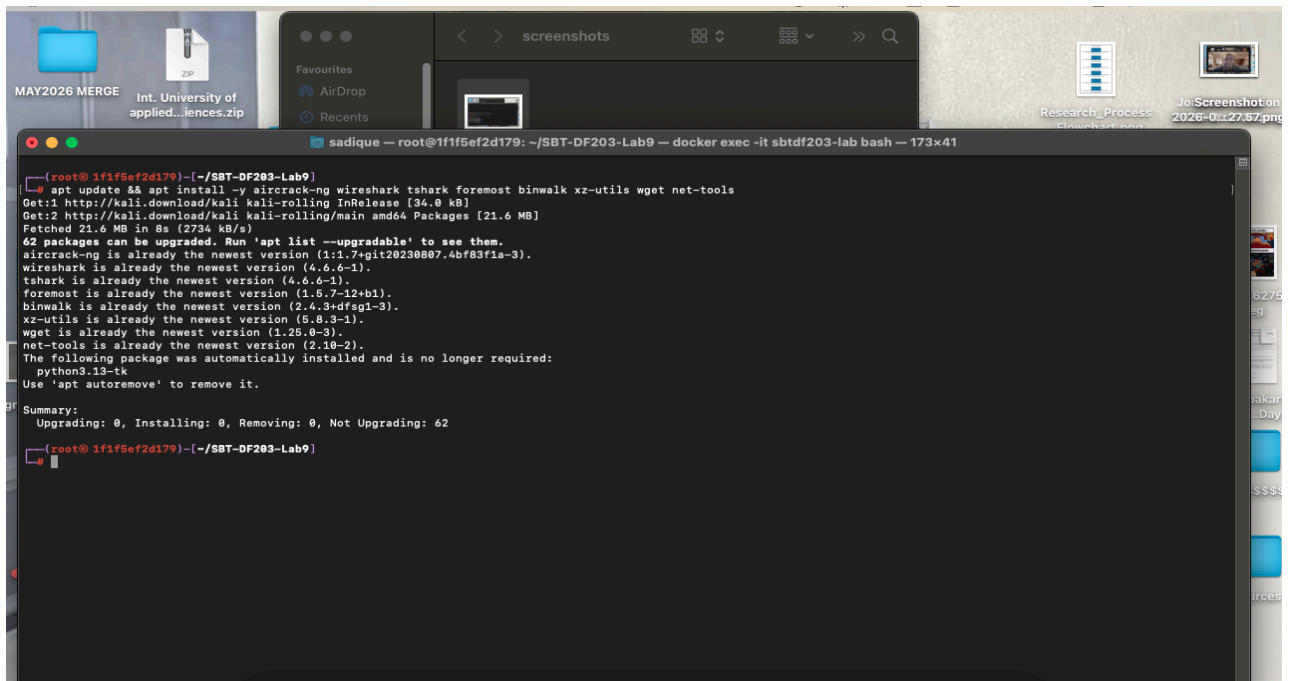
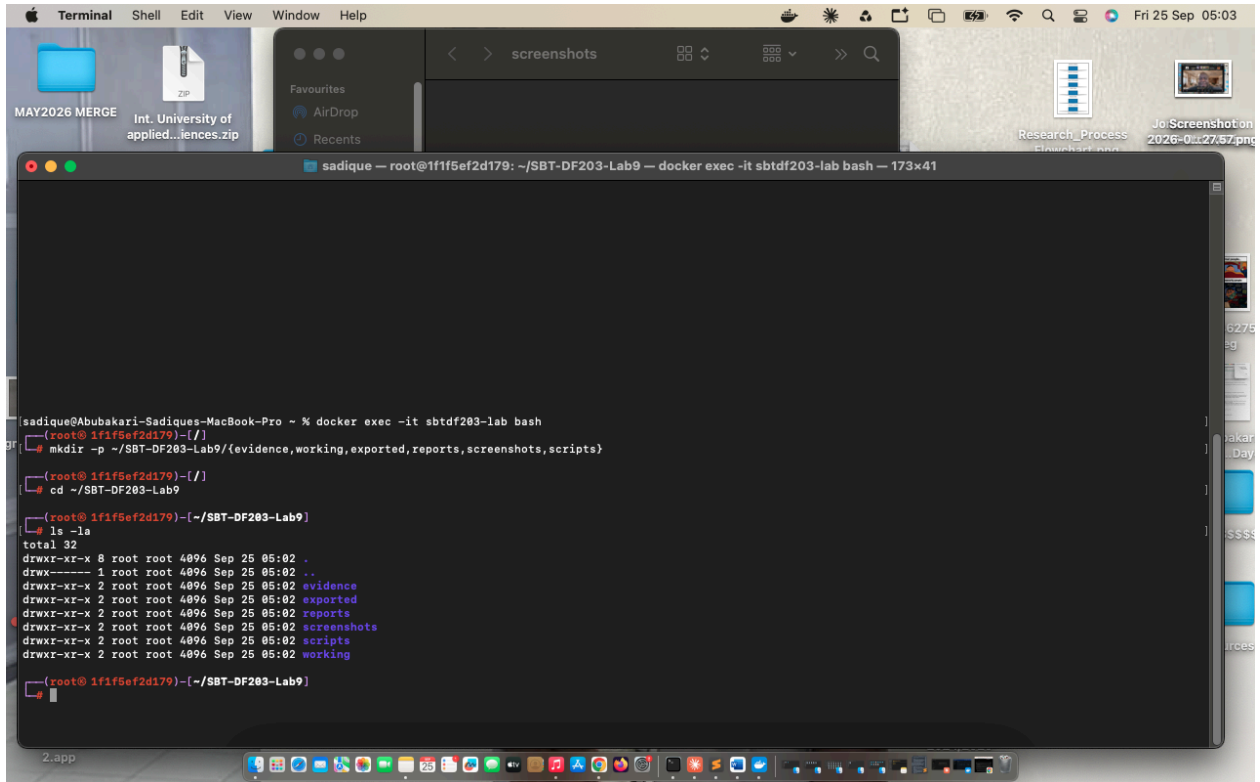
Item	Recommended Tool	Purpose
Operating System	Kali Linux Docker (sbtdf203-lab)	Analysis workstation.
Wireless Toolkit	aircrack-ng & airdecap-ng	Key recovery and offline WEP decryption.
Packet Analysis	capinfos, tshark, Wireshark	Protocol hierarchy dissection and endpoint inventory.
Carving Utilities	foremost, file	Extracting embedded media and file artifacts.
AI Assistant	Gemini	Troubleshooting command errors and organizing the report.
Evidence File	CodeGate CTF 2015 file.xz	Historical 802.11 WEP capture dataset.

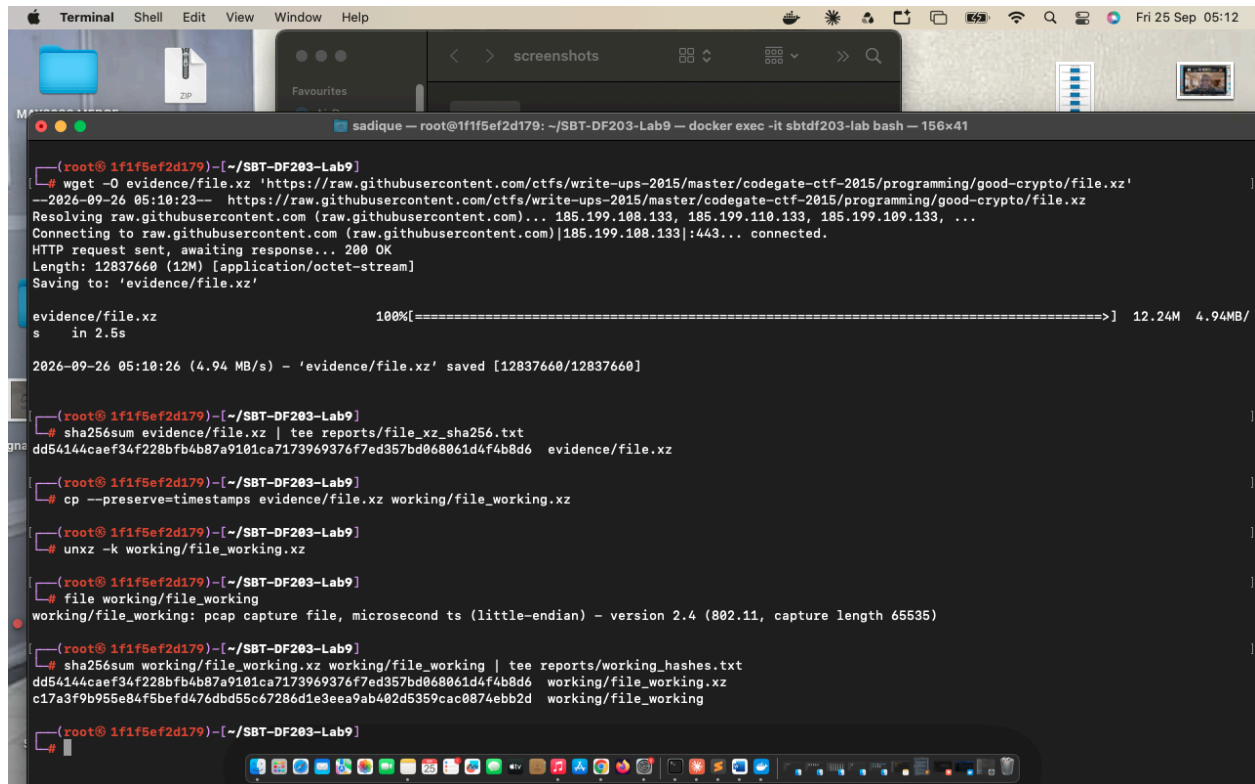
6. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/Lab Identifier	SBT-DF203-Lab9-AbubakariSadiqueHamidu
Trainee Name	Abubakari-Sadique Hamidu
Date Started	2026-09-25 05:10 UTC
Evidence File Name(s)	file.xz, file_working, file_working-dec
Original File.xz SHA-256	dd54144caef34f228bfb4b87a9101ca7173969 376f7ed357bd068061d4f4b8d6
Decompressed Capture SHA-256	c17a3f9b955e84f5befd476dbd55c67286d1e3 eea9ab402d5359cac0874ebb2d
Decrypted Capture SHA-256	167c91994c269777f9048227deb89882caf3cf 3c763977f2059604f9a6a40b04
Analysis Workstation	macOS Host / Kali Linux Docker Container (sbtdf203-lab)

7. Environment Initialization & Evidence Preparation

Actions Executed: Created standard workspace directories (evidence/, working/, reports/, screenshots/), installed required forensic tools (aircrack-ng, tshark, foremost, xz-utils), downloaded the historical CTF capture file.xz, recorded its SHA-256 checksum, decompressed it into file_working, and confirmed its 802.11 capture format.





```
(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# wget -O evidence/file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-09-26 05:10:23-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz          100%[=====] 12.24M  4.94MB/s
s in 2.5s

2026-09-26 05:10:26 (4.94 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# cp --preserve=timestamps evidence/file.xz working/file_working.xz

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# unxz -k working/file_working.xz

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (882.11, capture length 65535)

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
# sha256sum working/file_working.xz | tee reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3ee9ab402d5359cac0874ebb2d working/file_working

(root@ 1f1f5ef2d179) ~/SBT-DF203-Lab9
```

8. Part A & B – Wireless Capture Inventory & WEP IV Reuse Analysis

Actions Executed: Executed capinfos and tshark protocol hierarchy statistics to inventory 45,169 wireless frames. Identified Management (Type 0), Control (Type 1), and Data (Type 2) frame categories, extracted target BSSID 00:26:66:55:97:D6 (cgnetwork), and generated frequency summaries demonstrating widespread 24-bit Initialization Vector (IV) reuse across 15,713 protected WEP frames.

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x41

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# CAP=working/file_working

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# capinfos "$CAP" | tee reports/capinfos.txt
File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 07:51:54.364525
Latest packet time:  2015-03-06 07:56:27.334320
Data byte rate:     49 kBps
Data bit rate:      395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:             c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:               cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:  False
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
Capture length = 65535
Time precision = microseconds (6)
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 45169

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
#
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x41

Number of packets = 45169

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
Running as user "root" and group "root". This could be dangerous.

Protocol Hierarchy Statistics
Filter:

frame                                frames:45169 bytes:13511274
wlan                                frames:45169 bytes:13511274
data                                frames:15713 bytes:13104128

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# tshark -r "$CAP" -Y 'wlan' -T fields \
> -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
> | head -n 100 | tee reports/wlan_frame_sample.tsv
Running as user "root" and group "root". This could be dangerous.
1 2015-03-06T07:51:54.364526000+0000 1 11
2 2015-03-06T07:51:54.364527000+0000 1 11
3 2015-03-06T07:51:54.364525000+0000 1 11
4 2015-03-06T07:51:54.364526000+0000 1 11
5 2015-03-06T07:51:54.365550000+0000 1 11
6 2015-03-06T07:51:54.366063000+0000 1 11
7 2015-03-06T07:51:54.366062000+0000 1 11
8 2015-03-06T07:51:54.366061000+0000 1 11
9 2015-03-06T07:51:54.366063000+0000 1 11
10 2015-03-06T07:51:54.366061000+0000 1 11
11 2015-03-06T07:51:54.366064000+0000 1 11
12 2015-03-06T07:51:54.366063000+0000 1 11
13 2015-03-06T07:51:54.366063000+0000 1 11
14 2015-03-06T07:51:54.366063000+0000 1 11
15 2015-03-06T07:51:54.366064000+0000 1 11
16 2015-03-06T07:51:54.366062000+0000 1 11
17 2015-03-06T07:51:54.366575000+0000 1 11
18 2015-03-06T07:51:54.366575000+0000 1 11
19 2015-03-06T07:51:54.366573000+0000 1 11
20 2015-03-06T07:51:54
21 2015-03-06T07:51:54
```

```

Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdef203-lab bash — 156x41

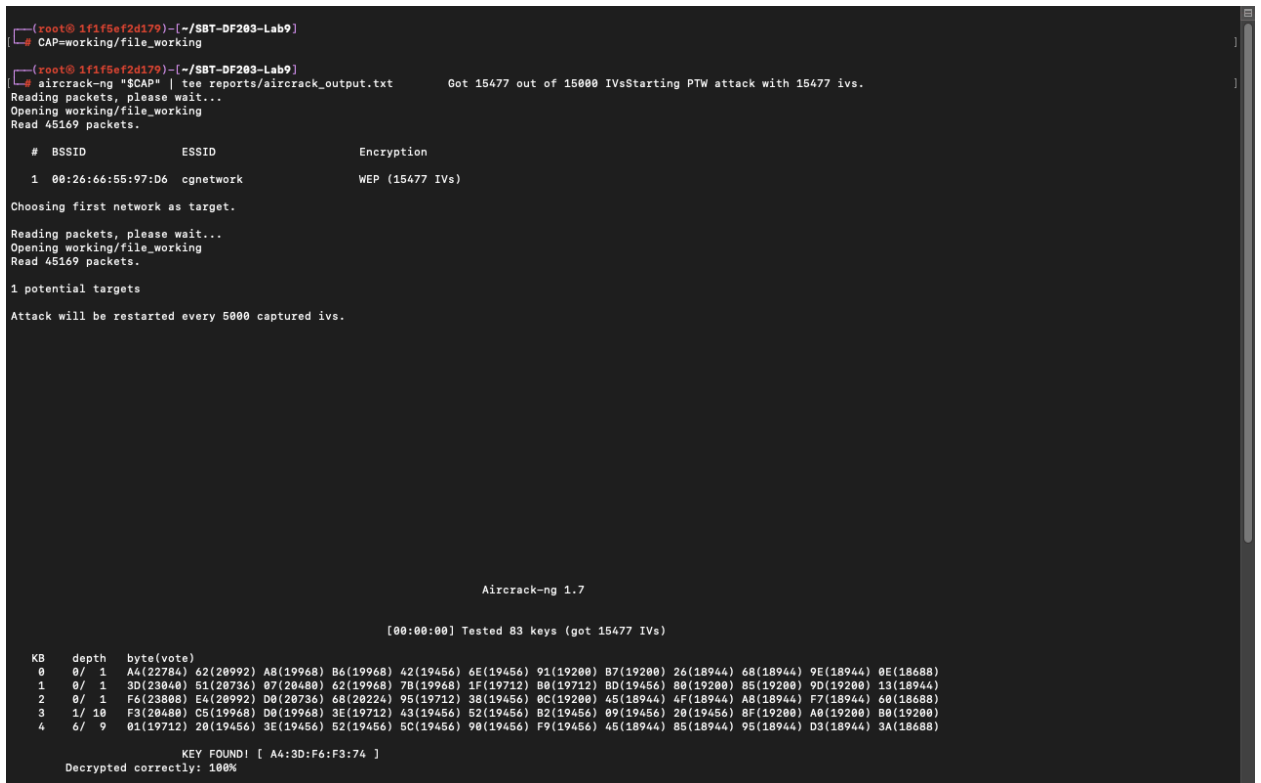
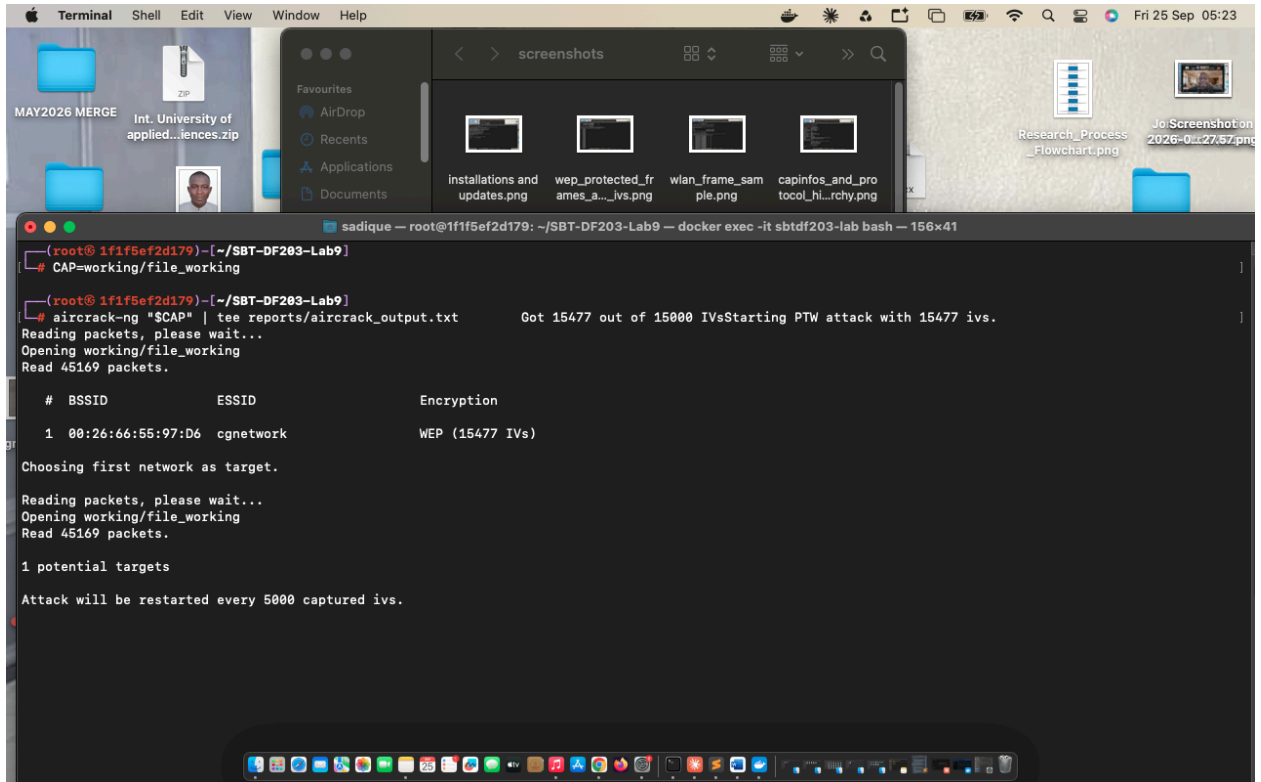
97 2015-03-06T07:51:54.381423000+0000 1 11
98 2015-03-06T07:51:54.382446000+0000 1 11
99 2015-03-06T07:51:54.382446000+0000 1 11
100 2015-03-06T07:51:54.382957000+0000 1 11

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# tshark -r "SCAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
> | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
Running as user "root" and group "root". This could be dangerous.
 7 0x42e8eb
 7 0x28d3eb
 6 0x95dfeb
 5 0x9415ed
 5 0x86e5eb
 5 0x7ae5eb
 5 0x28e7eb
 5 0x19e5eb
 4 0xd5d7eb
 4 0xd3e8eb
 4 0xc915ed
 4 0xc7d8eb
 4 0xade4eb
 4 0x73e5eb
 4 0x65d8eb
 4 0x4ad5eb
 4 0x31e7eb
 4 0x2723ed
 4 0x10e7eb
 3 0xffe5eb
 3 0xf609ed
 3 0xa307ed
 3 0x97dbeb
 3 0x96d1eb
 3 0x92e8eb
 3 0x91e8eb
 3 0x8adeeb
 3 0x630aed
 3 0x56e8eb
 3 0x3c1aed

```

9. Part C & D – Key Recovery & Offline Decryption

Actions Executed: Ran aircrack-ng using the PTW statistical attack against 15,477 collected IVs to recover the 40-bit WEP key A4:3D:F6:F3:74. Executed airdecap-ng with the recovered key to decrypt all WEP payload frames offline, generating file_working-dec with a 100% success rate and zero corrupted packets.



```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

KB  depth  byte(vote)
0    0/ 1    A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688)
1    0/ 1    3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) B0(19456) 80(19200) 85(19200) 9D(19200) 13(18944)
2    0/ 1    F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688)
3    1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200)
4    6/ 9    01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# cd working

(root@1f1f5ef2d179)~/SBT-DF203-Lab9/working
# airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets     0
Number of decrypted WPA packets     0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(root@1f1f5ef2d179)~/SBT-DF203-Lab9/working
# cd ..

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# cd working

(root@1f1f5ef2d179)~/SBT-DF203-Lab9/working
# airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets     0
Number of decrypted WPA packets     0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(root@1f1f5ef2d179)~/SBT-DF203-Lab9/working
# cd ..

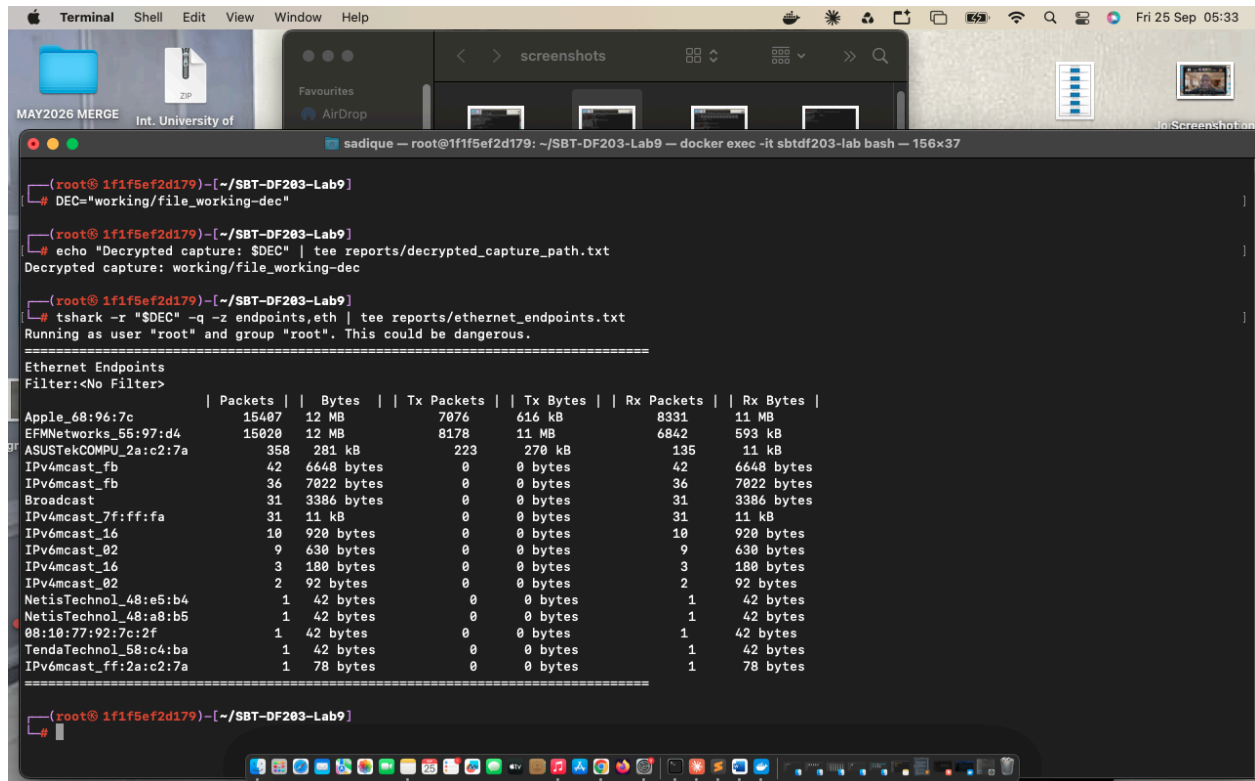
(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
209828 12540 -rw-r--r-- 1 root root 1283760 Sep 26 2026 working/file_working.xz
209830 13904 -rw-r--r-- 1 root root 14234002 Sep 26 2026 working/file_working
209851 12488 -rw-r--r-- 1 root root 12784501 Sep 25 05:26 working/file_working-dec

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3ee9ab402d5359cac0874ebb2d working/file_working
167c9199ac26977f904822deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
```

10. Part E & F – Endpoint Reconstruction & File Carving

Actions Executed: Inspected the decrypted traffic (file_working-dec) to build layer 3 IP/Ethernet endpoint lists and reconstruct TCP conversations. Exported transferred web content using tshark --export-objects http and carved embedded binary/media objects using foremost, establishing SHA-256 integrity hashes for all recovered file artifacts.



```
Terminal Shell Edit View Window Help
MAY2026 MERGE Int. University of
sadique -- root@1f1f5ef2d179: ~/SBT-DF203-Lab9 -- docker exec -it sbtdf203-lab bash -- 156x37

[root@1f1f5ef2d179]~/~/SBT-DF203-Lab9]
# DEC="working/file_working-dec"

[root@1f1f5ef2d179]~/~/SBT-DF203-Lab9]
# echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
Decrypted capture: working/file_working-dec

[root@1f1f5ef2d179]~/~/SBT-DF203-Lab9]
# tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
Running as user "root" and group "root". This could be dangerous.

Ethernet Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c 15407 12 MB 7076 616 kB 8331 11 MB
EFMNetworks_55:97:d4 15020 12 MB 8178 11 MB 6842 593 kB
ASUSTekCOMPU_2a:c2:7a 358 281 kB 223 270 kB 135 11 kB
IPv4mcast_fb 42 6648 bytes 0 0 bytes 42 6648 bytes
IPv4mcast_fb 36 7022 bytes 0 0 bytes 36 7022 bytes
Broadcast 31 3386 bytes 0 0 bytes 31 3386 bytes
IPv4mcast_7f:ff:fa 31 11 kB 0 0 bytes 31 11 kB
IPv4mcast_16 10 920 bytes 0 0 bytes 10 920 bytes
IPv4mcast_02 9 630 bytes 0 0 bytes 9 630 bytes
IPv4mcast_16 3 180 bytes 0 0 bytes 3 180 bytes
IPv4mcast_02 2 92 bytes 0 0 bytes 2 92 bytes
NetisTechnol_48:e5:b4 1 42 bytes 0 0 bytes 1 42 bytes
NetisTechnol_48:a8:b5 1 42 bytes 0 0 bytes 1 42 bytes
08:10:77:92:7c:2f 1 42 bytes 0 0 bytes 1 42 bytes
TendaTechnol_58:c4:ba 1 42 bytes 0 0 bytes 1 42 bytes
IPv4mcast_ff:2a:c2:7a 1 78 bytes 0 0 bytes 1 78 bytes

[root@1f1f5ef2d179]~/~/SBT-DF203-Lab9]
#
```

```
Terminal Shell Edit View Window Help
sadique -- root@1f1f5ef2d179: ~/SBT-DF203-Lab9 -- docker exec -it sbtdf203-lab bash -- 156x37

(root@1f1f5ef2d179)~/~/SBT-DF203-Lab9]
# tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
Running as user "root" and group "root". This could be dangerous.

IPv4 Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
192.168.0.15 15309 12 MB 6995 608 kB 8314 11 MB
198.90.20.111 8057 6916 kB 4446 6637 kB 3611 279 kB
199.27.79.193 5694 4825 kB 3131 4603 kB 2563 221 kB
192.168.0.9 329 277 kB 197 265 kB 132 11 kB
173.194.127.141 153 109 kB 80 102 kB 73 7364 bytes
173.194.127.55 143 89 kB 71 82 kB 72 6990 bytes
8.8.8.8 81 10 kB 40 6960 bytes 41 3391 bytes
199.16.158.186 66 15 kB 29 10 kB 37 5214 bytes
173.194.127.227 66 24 kB 26 15 kB 40 8455 bytes
173.194.127.229 55 22 kB 23 13 kB 32 9731 bytes
31.13.70.1 51 19 kB 26 15 kB 25 3151 bytes
173.241.248.220 44 6848 bytes 20 3122 bytes 24 3726 bytes
198.41.208.140 44 28 kB 21 25 kB 23 2718 bytes
224.0.0.251 42 6648 bytes 0 0 bytes 42 6648 bytes
173.194.127.237 36 9526 bytes 14 6435 bytes 22 3091 bytes
216.58.220.234 34 15 kB 16 12 kB 18 2497 bytes
192.168.0.1 33 12 kB 33 12 kB 0 0 bytes
173.241.248.219 33 5733 bytes 14 2807 bytes 19 2926 bytes
104.74.158.12 33 21 kB 18 20 kB 15 1236 bytes
74.125.203.156 32 20 kB 16 17 kB 16 2534 bytes
239.255.255.250 31 11 kB 0 0 bytes 31 11 kB
173.194.127.231 31 6816 bytes 12 5197 bytes 19 1610 bytes
198.41.208.137 28 16 kB 13 14 kB 15 1556 bytes
17.172.239.31 25 6504 bytes 12 3973 bytes 13 2531 bytes
173.194.72.95 23 4077 bytes 9 2232 bytes 14 1845 bytes
173.194.127.122 21 12 kB 11 11 kB 10 1759 bytes
69.59.169.185 18 5644 bytes 7 3675 bytes 11 1969 bytes
72.246.103.16 16 3697 bytes 6 1732 bytes 10 1965 bytes
23.23.127.192 15 3383 bytes 6 1370 bytes 9 2013 bytes

Zapp
```

```
Terminal Shell Edit View Window Help
sadique -- root@1f1f5ef2d179: ~/SBT-DF203-Lab9 -- docker exec -it sbtdf203-lab bash -- 156x37

10.0.0.1 1 60 bytes 1 60 bytes 0 0 bytes

(root@1f1f5ef2d179)~/~/SBT-DF203-Lab9]
# tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
Running as user "root" and group "root". This could be dangerous.

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
| 192.168.0.15:51007 | 1257 | 1885 kB | 978 | 73 kB | 2235 | 1958 kB | 113.588792000 | 45.4035 |
| 192.168.0.15:50959 | 1209 | 1816 kB | 959 | 71 kB | 2168 | 1887 kB | 21.074236000 | 34.8633 |
| 192.168.0.15:51008 | 1159 | 1739 kB | 947 | 70 kB | 2106 | 1810 kB | 113.588792000 | 46.0179 |
| 192.168.0.15:51001 | 1145 | 1725 kB | 927 | 66 kB | 2072 | 1791 kB | 63.066042000 | 26.6088 |
| 192.168.0.15:51009 | 687 | 1028 kB | 572 | 44 kB | 1259 | 1072 kB | 113.605175000 | 28.7983 |
| 192.168.0.15:51014 | 664 | 991 kB | 566 | 44 kB | 1230 | 1036 kB | 114.020474000 | 25.0538 |
| 192.168.0.15:51006 | 654 | 970 kB | 520 | 40 kB | 1174 | 1011 kB | 113.578039000 | 25.4942 |
| 192.168.0.15:51000 | 375 | 561 kB | 323 | 25 kB | 698 | 586 kB | 58.731642000 | 4.4552 |
| 192.168.0.15:50999 | 206 | 303 kB | 180 | 15 kB | 386 | 319 kB | 56.792636000 | 1.9390 |
| 192.168.0.15:51036 | 98 | 144 kB | 70 | 5705 bytes | 168 | 150 kB | 243.325633000 | 0.1777 |
| 192.168.0.15:50958 | 73 | 93 kB | 67 | 12 kB | 140 | 105 kB | 20.145401000 | 14.9846 |
| 192.168.0.15:51035 | 80 | 116 kB | 56 | 4756 bytes | 136 | 121 kB | 243.281602000 | 0.1398 |
| 192.168.0.15:50983 | 71 | 99 kB | 59 | 4392 bytes | 130 | 104 kB | 27.831547000 | 31.4188 |
| 192.168.0.15:50997 | 62 | 87 kB | 37 | 3826 bytes | 99 | 90 kB | 56.126972000 | 11.0492 |
| 192.168.0.15:51034 | 23 | 13 kB | 32 | 9731 bytes | 55 | 22 kB | 234.384001000 | 6.7568 |
| 192.168.0.15:51005 | 25 | 22 kB | 28 | 4757 bytes | 53 | 27 kB | 113.019956000 | 2.8111 |
| 192.168.0.15:50960 | 26 | 15 kB | 25 | 3151 bytes | 51 | 19 kB | 21.329724000 | 32.8418 |
| 192.168.0.15:51028 | 18 | 11 kB | 29 | 7603 bytes | 47 | 18 kB | 172.217100000 | 37.9452 |
| 192.168.0.15:51003 | 21 | 25 kB | 23 | 2718 bytes | 44 | 28 kB | 102.780857000 | 32.0878 |
| 192.168.0.15:51018 | 20 | 24 kB | 16 | 1494 bytes | 36 | 26 kB | 115.102391000 | 30.4286 |
| 192.168.0.15:50973 | 15 | 5156 bytes | 19 | 2640 bytes | 34 | 7796 bytes | 25.665085000 | 31.5233 |
| 192.168.0.15:51002 | 18 | 20 kB | 15 | 1236 bytes | 33 | 21 kB | 63.175099000 | 30.4746 |
| 192.168.0.15:50974 | 14 | 5082 bytes | 18 | 2574 bytes | 32 | 7656 bytes | 25.665085000 | 31.5223 |
| 192.168.0.15:51016 | 13 | 15 kB | 19 | 1749 bytes | 32 | 16 kB | 115.101879000 | 30.4439 |
| 192.168.0.15:51020 | 16 | 17 kB | 16 | 2534 bytes | 32 | 20 kB | 115.146935000 | 31.5115 |
| 192.168.0.15:50956 | 13 | 14 kB | 15 | 1556 bytes | 28 | 16 kB | 13.669693000 | 31.2704 |

Zapp
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
> | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
Running as user "root" and group "root". This could be dangerous.
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TCP
43 17.172.239.31 192.168.0.15 TLSv1
44 17.172.239.31 192.168.0.15 TLSv1
45 17.172.239.31 192.168.0.15 TLSv1
47 0.0.0.0 255.255.255.255 DHCP
48 0.0.0.0 255.255.255.255 DHCP
49 192.168.0.15 17.172.239.31 TCP
50 192.168.0.15 17.172.239.31 TCP
51 192.168.0.1 192.168.0.15 DHCP
52 192.168.0.15 17.172.239.31 TLSv1
55 17.172.239.31 192.168.0.15 TCP
56 192.168.0.15 17.172.239.31 TLSv1
57 17.172.239.31 192.168.0.15 TCP
58 17.172.239.31 192.168.0.15 TLSv1
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
mkdir -p exported/http_objects exported/foremost

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
Running as user "root" and group "root". This could be dangerous.
1 0.000000 EFMNetworks 55:97:d4 → Broadcast ARP 42 Who has 192.168.0.15? Tell 192.168.0.1
2 0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.168.1.41
3 0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
4 0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? Tell 192.168.1.12
5 0.944186 Apple_68:96:7c → EFMNetworks 55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
6 0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.1? Tell 192.168.10.113
7 0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
8 0.944784 EFMNetworks 55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
9 0.947778 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
10 0.949386 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6be1c
12 0.954937 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
13 0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
14 0.956994 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
15 0.959034 Apple_68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
16 0.960066 Apple_68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
17 1.026186 Apple_68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
18 1.029698 Apple_68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
19 1.029696 EFMNetworks 55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
20 1.030202 192.168.0.15 → 8.8.8.8 DNS 73 Standard query 0x9583 A www.apple.com
21 1.036368 192.168.0.15 → 8.8.8.8 DNS 85 Standard query 0xbe40 A 48-courier.push.apple.com
22 1.187392 192.168.0.15 → 224.0.0.2 IGMPv2 46 Leave Group 224.0.0.251
23 1.187392 192.168.0.15 → 224.0.0.251 IGMPv2 46 Membership Report group 224.0.0.251
24 1.187984 192.168.0.15 → 224.0.0.2 IGMPv2 46 Leave Group 224.0.0.251
25 1.189439 192.168.0.15 → 224.0.0.251 IGMPv2 46 Membership Report group 224.0.0.251
26 1.350208 8.8.8.8 → 192.168.0.15 DNS 203 Standard query response 0x9583 A www.apple.com CNAME www.isg-apple.com.akadns.net CNAME www.apple.com.e
key.net CNAME e3191.dscc.akamaiedge.net A 184.74.154.228
27 1.350718 8.8.8.8 → 192.168.0.15 DNS 299 Standard query response 0xbe40 A 48-courier.push.apple.com CNAME 48.courier-push-apple.com.akadns.net C
E us-courier.push-apple.com.akadns.net A 17.172.239.31 A 17.172.239.127 A 17.172.232.188 A 17.172.239.83 A 17.172.232.166 A 17.172.239.25 A 17.172.232.17
A 17.172.232.185
```

```
Terminal Shell Edit View Window Help
MAY2026 MERGE Int. University of applied...iences.zip
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
Processing: working/file_working-dec
[*]

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/http_objects/figure-4.png: PNG image data, 440 x 173, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-print.png: PNG image data, 65 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/analytics%3Fconly=true&visitor_id=73685114&api_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20Zeu%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2Ffr%2Fnetsec: ASCII text
exported/http_objects/qqy-UiLNKP-Vf0dbcs6r6-vvDin1pK8aKteLpeZ5c0A.woff: Web Open Font Format, TrueType, length 12768, version 1.1
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3A%222sesionId%22%3A%224d0d205eea8f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(4).com%2F8iVy7t6%22%27D: ASCII text, with no line terminat
ors
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3A%222sesionId%22%3A%224d0d205eea8f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(1).com%2F8iVy7t6%22%27D: ASCII text, with no line terminat
ors
exported/http_objects/upw-theme-standard.min.css: ASCII text, with very long lines (1018), with no line terminators
exported/http_objects/bootstrap-responsive.min.css: ASCII text, with very long lines (16608)
exported/http_objects/ogli5PA(1).mp4: data
exported/http_objects/sharePlayer.js%3f1425580383: ASCII text, with very long lines (2499)
exported/http_objects/v2(1): JSON text data
exported/http_objects/trackalyze.js: ASCII text, with CRLF line terminators
exported/http_objects/heart.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-facebook.png: PNG image data, 95 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/blah.css: ASCII text
exported/http_objects/ogli5PA.mp4: data
exported/http_objects/b9QBgL0iMZfDspmcXcE8nPZRQubYfRc-1FWT9T9EE8M.woff: Web Open Font Format, TrueType, length 13324, version 1.1
exported/http_objects/figure-16.png: PNG image data, 230 x 144, 8-bit/color RGBA, non-interlaced
exported/http_objects/analytics%3Fver=3&visitor_id=&api_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20Zeu%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer=http%3A%2.com%2Ffr%2Fnetsec: ASCII text
exported/http_objects/jquery-ui.css: ASCII text, with very long lines (551)
exported/http_objects/object%26524: data
exported/http_objects/jquery-1.10.2.min.js: JavaScript source, ASCII text, with very long lines (32072)
exported/http_objects/category-strip.png: PNG image data, 16 x 20, 8-bit/color RGBA, non-interlaced
exported/http_objects/css%3Ffamily=Oswald%7C: ASCII text
exported/http_objects/twitter.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/portfolio-slideshow.min.css: ASCII text, with very long lines (600)
exported/http_objects/share-linkedin.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/ogli5PA.gifv: HTML document, ASCII text
```

```
Terminal Shell Edit View Window Help
MAY2026 MERGE Int. University of applied...iences.zip
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab9 — docker exec -it sbtdf203-lab bash — 156x37

(root@1f1f5ef2d179)~/SBT-DF203-Lab9
# find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/http_objects/figure-4.png: PNG image data, 440 x 173, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-print.png: PNG image data, 65 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/analytics%3Fconly=true&visitor_id=73685114&api_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20Zeu%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2Ffr%2Fnetsec: ASCII text
exported/http_objects/qqy-UiLNKP-Vf0dbcs6r6-vvDin1pK8aKteLpeZ5c0A.woff: Web Open Font Format, TrueType, length 12768, version 1.1
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3A%222sesionId%22%3A%224d0d205eea8f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(4).com%2F8iVy7t6%22%27D: ASCII text, with no line terminat
ors
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3A%222sesionId%22%3A%224d0d205eea8f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(1).com%2F8iVy7t6%22%27D: ASCII text, with no line terminat
ors
exported/http_objects/upw-theme-standard.min.css: ASCII text, with very long lines (1018), with no line terminators
exported/http_objects/bootstrap-responsive.min.css: ASCII text, with very long lines (16608)
exported/http_objects/ogli5PA(1).mp4: data
exported/http_objects/sharePlayer.js%3f1425580383: ASCII text, with very long lines (2499)
exported/http_objects/v2(1): JSON text data
exported/http_objects/trackalyze.js: ASCII text, with CRLF line terminators
exported/http_objects/heart.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-facebook.png: PNG image data, 95 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/blah.css: ASCII text
exported/http_objects/ogli5PA.mp4: data
exported/http_objects/b9QBgL0iMZfDspmcXcE8nPZRQubYfRc-1FWT9T9EE8M.woff: Web Open Font Format, TrueType, length 13324, version 1.1
exported/http_objects/figure-16.png: PNG image data, 230 x 144, 8-bit/color RGBA, non-interlaced
exported/http_objects/analytics%3Fver=3&visitor_id=&api_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20Zeu%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer=http%3A%2.com%2Ffr%2Fnetsec: ASCII text
exported/http_objects/jquery-ui.css: ASCII text, with very long lines (551)
exported/http_objects/object%26524: data
exported/http_objects/jquery-1.10.2.min.js: JavaScript source, ASCII text, with very long lines (32072)
exported/http_objects/category-strip.png: PNG image data, 16 x 20, 8-bit/color RGBA, non-interlaced
exported/http_objects/css%3Ffamily=Oswald%7C: ASCII text
exported/http_objects/twitter.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/portfolio-slideshow.min.css: ASCII text, with very long lines (600)
exported/http_objects/share-linkedin.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/ogli5PA.gifv: HTML document, ASCII text
```

11. Required Findings Worksheet

Question / Field	Finding
Capture Format / Duration	pcap (IEEE 802.11), 272.97 seconds
BSSID / ESSID	00:26:66:55:97:D6 / cgnetwork
Protected Frame Count	15,713 frames
Repeated IV Evidence	Multiple IVs repeated up to 7 times (e.g., 0x42e8eb)
Recovered WEP40 Key	A4:3D:F6:F3:74
Decrypted Capture Name	file_working-dec
Decrypted Capture SHA-256	167c91994c269777f9048227deb89882caf3cf 3c763977f2059604f9a6a40b04
Security Conclusion	WEP is fundamentally insecure due to 24-bit IV reuse and RC4 keystream exposure; upgrade to WPA2/WPA3.

12. WEP Cryptographic Weakness & Defensive Recommendations

- **Short 24-bit Initialization Vector:** Causes inevitable IV collisions on busy networks, allowing attackers to XOR ciphertexts and recover plaintexts.
- **Weak CRC-32 ICV:** Provides no cryptographic integrity protection, enabling packet tampering and bit-flipping attacks.

- **Defensive Recommendation:** Retire WEP entirely in favor of WPA2-AES (CCMP) or WPA3 with Protected Management Frames (PMF).

Academic Integrity and Authorization Statement

Student Declaration: I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Signed: Abubakari-Sadique Hamidu

Student ID: 2025-FWSD-11392

Date: 25 September 2026